Stochastic Dynamics in Phytoplankton Layers

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Phytoplankton Dynamics

Reference Work [1]:
F. M. Ventrella et al., *Microswimmer trapping in surface waves with shear*, Proc. R. Soc. A 479 (2023).

We focus on the **Pure Shear** regime.

Governing Equations:

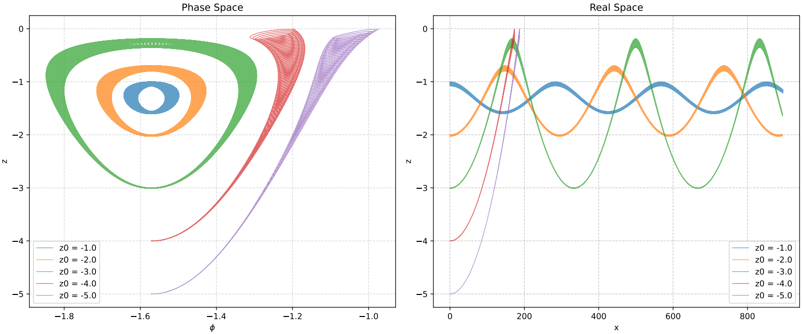
$$\dot{x} = \underbrace{\alpha e^z \cos(x - t)}_{\text{Wave Transport}} + \underbrace{\sigma(\beta + z)}_{\text{Shear Flow}} + \underbrace{\nu \sin\phi}_{\text{Swimming}}$$

$$\dot{z} = \underbrace{\alpha e^z \sin(x - t)}_{\text{Wave Transport}} + \underbrace{\nu \cos\phi}_{\text{Swimming}}$$

$$\dot{\phi} = \underbrace{\lambda\alpha e^z \cos(x - t + 2\phi)}_{\text{Rotation by Waves}} + \underbrace{\frac{\sigma}{2}(1 + \lambda\cos 2\phi)}_{\text{Rotation by Shear}}$$

Parameters: α *(wave steepness),* σ *(shear rate),* ν *(swimming speed),* λ *(shape).*

Deterministic Baseline: Trapping



Phase space and real space trajectories (Reproduced)

- **Method:** Equations integrated via **RK4**, discarding unphysical paths ($z > 0$).
- **Result:** Swimmers are trapped in **closed orbits** around a neutral equilibrium point.

Real-World Context

Do these layers actually exist?

San Diego, California (2020) [2]

New York Bight (2015) [3]

- **Reality vs. Model:** Observed thin layers are subject to biological variability, unlike perfect deterministic trajectories.
- **Goal:** We extend the model with stochastic terms to capture these realistic fluctuations.

Introducing Stochasticity

Model Extension: Deterministic → **Stochastic Differential Equation (SDE)**

Why Noise in ϕ?

- **Biological Source:** Imperfect motility / Flagellar asymmetry.
- **Kinematic Coupling:** Velocities $\dot{x}$ and $\dot{z}$ depend explicitly on ϕ.
  - Noise in orientation directly drives spatial fluctuations.
  - Small $\delta\phi$ errors accumulate into large trajectory deviations.

SDE for ϕ:

$$d\phi = \underbrace{[\dots]}_{\text{Deterministic Torque}} dt + \underbrace{\sqrt{2D_r}\, dW_t}_{\text{Rotational Noise}}$$

Numerical Method: Euler-Maruyama ($dt = 0.1$).

Stochastic Dynamics: Long-Time Behavior

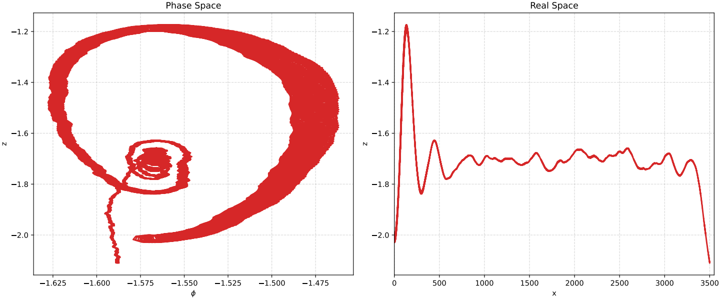


Figure 1: Averaged of the simulated paths $T = 4 \times 10^{-4}$ ($D_r = 10^{-5}$)

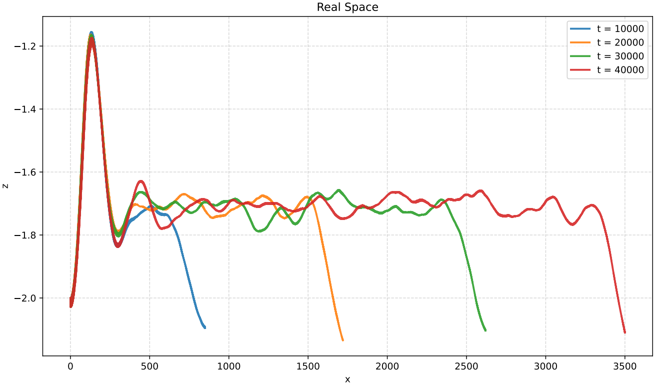
1. Persistence of Structure

- Accumulation layer survives noise.
- Trajectories "smeared" but trapped.

2. Downward Drift

- No closed orbits (vs. deterministic).
- Escape from boundary.

Time Evolution: The "Leaking" Layer



Key Observations ($t_1 \to t_4$):

- **Population Decay:** Plankton escaping from the layer.
- **Conclusion:** Thin layers are **transient / metastable** structures.

1. The Deterministic Model
2. Introducing Stochasticity
3. Numerical Schemes for Fokker Planck
    1. Fokker Planck Theory
    2. Discretisation
    3. Stability
    4. Chang-Cooper Scheme
4. Results & Conclusions
5. Appendix

From SDE to Fokker-Planck

- **1. Stochastic Differential Equation (Single Trajectory)**

$$\frac{dx}{dt} = \mu(x) + \sqrt{2D} \cdot \xi(t)$$

$\downarrow$ *Evolution of the probability density $p(x,t)$ [4]*

- **2. Fokker-Planck Equation (Ensemble)**

$$\frac{\partial p}{\partial t} = \underbrace{-\frac{\partial}{\partial x}\left[\mu(x)p\right]}_{\text{Advection (Transport)}} + \underbrace{\frac{1}{2}\frac{\partial^2}{\partial x^2}\left[\sigma^2(x)p\right]}_{\text{Diffusion (Spreading)}}$$

Numerical Strategy: Operator Splitting

The Full Fokker–Planck Equation

The PDE involves coupled advection (μ) and diffusion (D_r):

$$\partial_t f = -\partial_x\big(\mu_x f\big) - \partial_z\big(\mu_z f\big) - \partial_\phi\big(\mu_\phi f\big) + D_r\,\partial_\phi^2 f \;\equiv\; \mathcal{L}f\,.$$

The Solution: Operator Splitting

We decouple the full operator $\mathcal{L}$ into two simpler sub-problems:

$$\mathcal{L} = \mathcal{L}_{\text{x, adv}} + \mathcal{L}_{\text{z, adv}} + \mathcal{L}_{\phi,\text{ adv}} + \mathcal{L}_{\phi,\text{ diff}}$$

Numerical Discretization: Time Stepping Strategies

1. Semi-Implicit Time Stepping (Crank-Nicolson)

We treat diffusion implicitly (Crank-Nicolson) while using Upwind discretization for all advection terms (including ϕ).

$$\frac{f^{n+1} - f^n}{\Delta t} = \underbrace{(\mathcal{L}_{\text{x, adv}} + \mathcal{L}_{\text{z, adv}} + \mathcal{L}_{\phi,\text{adv}})f^n}_{\substack{\text{All Advection} \\ (x,z,\phi)}} + \underbrace{\frac{1}{2}\mathcal{L}_{\phi,\text{diff}}(f^{n+1} + f^n)}_{\text{Crank-Nicolson}}$$

- **Advection:** All handled via **Upwind** schemes.

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2. Fully Explicit Stepping (Chang-Cooper)

The entire equation is advanced using Explicit Euler. The angular part is treated with the Chang-Cooper scheme.

$$f^{n+1} = f^n + \Delta t \left(\underbrace{(\mathcal{L}_{\text{x, adv}} + \mathcal{L}_{\text{z, adv}})f^n}_{\text{Spatial}} + \underbrace{(\mathcal{L}_{\phi,\text{adv}} + \mathcal{L}_{\phi,\text{diff}})f^n}_{\text{Angular}} \right)$$

Stability: Crank–Nicolson / Chang–Cooper + CFL for drift

- **Diffusion in ϕ (implicit Crank–Nicolson)**

  The CN step is **unconditionally stable** for the diffusion term
- **Explicit solvers**

  The stability of the explicit step is controlled by the usual CFL condition. We monitor the Courant numbers

$$C_x = \max|\mu_x| \frac{\Delta t}{\Delta x}, \quad C_z = \max|\mu_z| \frac{\Delta t}{\Delta z},$$

$$C_\phi = \max|\mu_\phi| \frac{\Delta t}{\Delta \phi} + \underbrace{\frac{D_r}{\Delta \phi^2}}_{\text{Chang-Cooper only}}$$

and require

$$\Delta t \lesssim \frac{1}{C_x + C_z + C_\phi}$$

for stability of the explicit update.

ϕ–Direction Flux: Chang–Cooper Equation

- **The Chang–Cooper Flux (J_ϕ) :**
  Combines weighted drift and central diffusion [5]

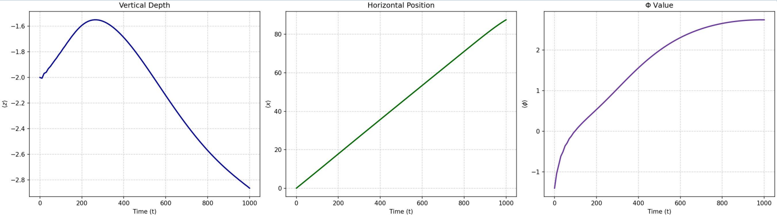
$$J_{\phi,k+\frac{1}{2}} = \underbrace{\mu_{\phi,k+\frac{1}{2}}\left[(1-\delta)f_{k+1} + \delta f_k\right]}_{\text{Drift (Upwinded)}} - \underbrace{D_{k+\frac{1}{2}}\frac{f_{k+1}-f_k}{\Delta\phi}}_{\text{Diffusion (Central Difference)}} .$$

- **Divergence in ϕ ($\partial_\phi J_\phi$):**
  The rate of change in the density at cell k

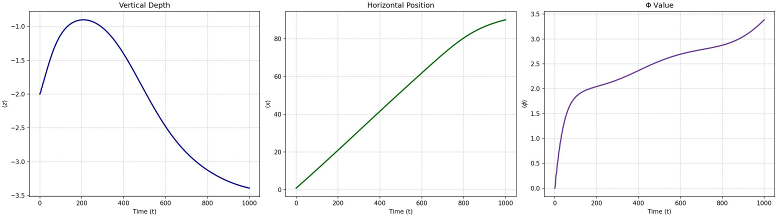
$$(\partial_\phi J_\phi)_k = -\frac{J_{\phi,k+\frac{1}{2}} - J_{\phi,k-\frac{1}{2}}}{\Delta\phi}.$$

- **The overall scheme preserves positivity and the analytical steady-state solution.**

Time Series Comparison of the Averages ($D_r = 10^{-5}$)

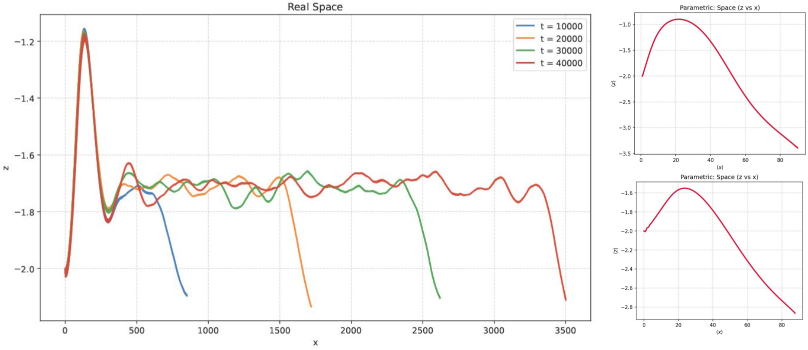


Chang-Cooper



Crank-Nicolson

Trajectory Comparison



(left) Stochastic paths, (upper right) FP Crank-Nicolson, (lower right) FP Chang Cooper

Conclusion

1. **Chang-Cooper** $\succ$ **Crank-Nicolson**:
   - Theoretically superior for stability.
   - Guaranteed positivity for high-drift regimes ($D_r \approx 10^{-5}$).
2. **SDE** $\gg$ **Fokker-Planck**
   - Better transparency.
   - Enabled direct and better global understanding
3. **The Limitation of Fokker-Planck**
   - No Steady State
   - FP became an expensive tool to track simple mass loss.
4. **Implementation** $\gg$ **Algorithm**:
   - **Numba (JIT)** was the decisive factor for speed.

Outlook

1. **Quantify Exit Rates**
   - Confirm residence time statistics and exponential decay.
2. **Higher Moment Analysis**
   - Compare variance and skewness, not just the mean path.
3. **Dimensional Analysis**
   - Reduce parameters.
4. **High-Performance Computing**
   - Scale to "Ultra-Sharp" grids to eliminate potential numerical heating.
5. **Experimental Validation**
   - Benchmark model against real micro-swimmer data.

Thank You!

Work Distribution

- **Pietro Riccardi**:
  Simulation and Analysis of Stochastic Differential Equations (SDEs)
- **Filip Baran**:
  Implementation of Crank-Nicolson for the Fokker-Planck Equation
- **Jasper Aden**:
  Implementation of Chang-Cooper Scheme for the Fokker-Planck Equation

[1] F. M. Ventrella, N. Pujara, G. Boffetta, M. Cencini, J.-L. Thiffeault, and F. De Lillo.
Microswimmer trapping in surface waves with shear.
Proc. R. Soc. A, 479:20230280, 2023.

[2] Scripps Institution of Oceanography.
Red tide at torrey pines state beach.
Photograph, 2020.
University of California San Diego.

[3] NASA Earth Observatory.
Algal bloom off new york and new jersey.
Satellite Image (Landsat 8), 2015.
National Aeronautics and Space Administration.

[4] Hannes Risken.
The Fokker-Planck Equation: Methods of Solution and Applications, volume 18 of *Springer Series in Synergetics*.

Springer-Verlag, Berlin, Heidelberg, 2nd edition, 1989.

[5] J. S. Chang and G. Cooper.
A practical difference scheme for fokker–planck equations.
Journal of Computational Physics, 6(2):1–16, 1970.

The Stochastic Process

- **General SDE:**

$$dX_t = \underbrace{\mu(X_t, t)dt}_{\text{Deterministic Drift}} + \underbrace{\sigma(X_t, t)dW_t}_{\text{Stochastic Diffusion}}$$

  - X_t: State variable (e.g., position).
  - dW_t: Wiener increment ($\mathbb{E}[dW_t] = 0, \mathbb{E}[dW_t^2] = dt$).

- **Comparison to Langevin Equation (Overdamped):**

$$\frac{dx}{dt} = -\frac{1}{\gamma}\nabla V(x) + \sqrt{2D} \cdot \xi(t)$$

$$\begin{array}{rcll} \textbf{Drift } \mu(x) & \longleftrightarrow & -\frac{1}{\gamma}\nabla V(x) & \text{(Force/Friction)} \\ \textbf{Diffusion } \sigma(x) & \longleftrightarrow & \sqrt{2D} & \text{(Noise Intensity)} \\ \textbf{Noise } dW_t/dt & \longleftrightarrow & \xi(t) & \text{(White noise)} \end{array}$$

The Bridge: Evolution of an Expectation

We track how an arbitrary test function $f(X_t)$ evolves over time.

- **Taylor Expansion (Second Order):**
  Substituting $dX_t = \mu dt + \sigma dW_t$:

$$df = f'(X_t)[\mu dt + \sigma dW_t] + \frac{1}{2} f''(X_t)[\underbrace{\sigma^2 (dW_t)^2}_{dt}]$$

- **Taking the Expectation:**

$$\frac{d}{dt}\mathbb{E}[f(X_t)] = \mathbb{E}\left[\mu(X_t)\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2(X_t)\frac{\partial^2 f}{\partial x^2}\right]$$

We must keep 2nd order terms because $dW_t \sim \sqrt{dt}$, so $(dW_t)^2 \sim dt$.

The Result: The Fokker-Planck Equation

How do we get the density $p(x,t)$?

1. **Integral Form:**

$$\int f(x)\frac{\partial p}{\partial t}dx = \int \left(\mu(x)\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2(x)\frac{\partial^2 f}{\partial x^2}\right) p(x,t)dx$$

2. **Integration by Parts:**

$$\int f(x)\frac{\partial p}{\partial t}dx = \int f(x)\left(-\frac{\partial}{\partial x}[\mu p] + \frac{1}{2}\frac{\partial^2}{\partial x^2}[\sigma^2 p]\right) dx$$

3. **Match Integrands:**

Fokker-Planck Equation

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x}[\mu(x,t)p] + \frac{1}{2}\frac{\partial^2}{\partial x^2}[\sigma^2(x,t)p]$$

ϕ–Direction Flux: Chang–Cooper Parameters

- **Face Drift (μ) and Diffusion (D):**
  Averaged to the cell face $k + \frac{1}{2}$

$$\mu_{\phi,k+\frac{1}{2}} = \frac{1}{2}\big(\mu_{\phi,k} + \mu_{\phi,k+1}\big), \qquad D_{k+\frac{1}{2}} = \frac{1}{2}\big(D_k + D_{k+1}\big).$$

- **Péclet Number (Pe):**
  Ratio of advective drift to diffusion across the cell

$$\mathrm{Pe}_{k+\frac{1}{2}} = \frac{\mu_{\phi,k+\frac{1}{2}}\,\Delta\phi}{D_{k+\frac{1}{2}}}.$$

- **Chang–Cooper Weight (δ):**
  Upwinding factor required for positivity and correct equilibrium

$$\delta(\mathrm{Pe}) = \frac{1}{\mathrm{Pe}} - \frac{1}{e^{\mathrm{Pe}} - 1}.$$